

Climbing-specific finger flexor performance and forearm muscle oxygenation in elite male and female sport climbers

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Abstract Climbing performance relies to a great extent on the performance of the finger flexor muscles. Only a few studies investigated this performance in top class climbers and only one study compared gender-specific differences. This study investigated the climbing-specific finger flexor strength and endurance and related muscular oxygenation in 12 elite female and male climbers and 12 non-climbers. After the assessment of maximum voluntary finger flexor contraction (MVC), two isometric finger flexor endurance tests were performed at 40% MVC until exhaustion. A continuous isometric test was followed by an intermittent isometric test (10 s contraction, 3 s rest). Changes in oxygenation of finger flexor muscles were recorded using near infrared spectroscopy. MVC and strength-to-weight ratio were greater in climbers than non-climbers ($P = 0.003$; $P < 0.001$) and greater in men than women ($P < 0.001$; $P = 0.002$). Time to task failure for the intermittent test and

the force-time integrals for the continuous and the intermittent test were also significantly greater in climbers ($P = 0.030$; $P = 0.027$; $P = 0.005$). During the intermittent test, re-oxygenation of the working muscles was faster in climbers ($P < 0.05$) without gender-specific differences. Close correlations were demonstrated between the best on-sight climbing performance and strength-to-weight ratio ($r^2 = 0.946$, $P < 0.001$) only in female climbers. The superior intermittent finger flexor endurance of climbers over non-climbers may be explained by the faster re-oxygenation of the finger flexor muscles during the short rest phases.

Keywords Climbing · Finger flexor muscles · Strength · Isometric · Endurance · Oxygenation

Introduction

The development of artificial indoor climbing walls made sport climbing accessible for a broad public all year and the number of climbing walls is continuously growing throughout Europe. Indoor climbing is considered a low-risk sport (Schöffl et al. 2010). These facts undoubtedly contributed to the steeply increasing number of young elite sport climbers. The fast evolution in sport climbing put the scientific focus on potential determinants of climbing performance. Male and female elite climbers tend to be relatively small, with a low body mass and a low percentage of body fat (España-Romero et al. 2009; Michailov et al. 2009; Morrison and Schöffl 2007; Giles et al. 2006; Watts 2004; Watts et al. 2003; Grant et al. 2001; Mermier et al. 2000). The aerobic capacity seems not to be a limiting factor of climbing performance (Watts 2004; Sheel et al. 2003; Booth et al. 1999; Watts and Drobish 1998; Watts

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et al. 1996; Billat et al. 1995). Actually, only 42% of the individual running VO_2max has been reported to be used during hard rock climbing (Billat et al. 1995). However, heart rates have been shown to be disproportionately high compared to the individual VO_2 during climbing (Sheel et al. 2003; Mermier et al. 1997) and blood lactate concentrations after sub maximal climbing indicated exercise intensities above the individual lactate threshold (Booth et al. 1999). All these observations suggest that anaerobic energy-yielding metabolic processes play an important role during climbing.

Performance of finger flexor muscles in climbers

In contrast to many other sport disciplines, climbing performance relies to a great extent on the performance of small muscle groups, especially the finger flexor muscles. Situations raising the stress on finger flexor muscles by enhancing climbing difficulty, climbing angle and climbing time lead to increased lactate production by those muscles (Schöffl et al. 2006; Noé et al. 2001; Booth et al. 1999; Watts and Drobish 1998; Watts et al. 1996; Billat et al. 1995). The metabolic acidosis in muscles involved in gripping is considered a main factor of “climbing fatigue” (Morrison and Schöffl 2007; Giles et al. 2006; Schöffl et al. 2006; Sheel 2004; Wall et al. 2004; Booth et al. 1999; Watts et al. 1996). In competitive bouldering, the exercise-to-recovery ratio was 13:1 (White and Olsen 2010). This constant ratio between rather long high intensity finger gripping tasks and short rest phases demonstrates the outstanding role of rapid recovery during rest phases. Besides, finger flexor strength and finger flexor endurance are key elements of climbing performance (España-Romero et al. 2009; MacLeod et al. 2007; Vigouroux and Quaine 2006; Wall et al. 2004; Quaine et al. 2003; Mermier et al. 2000). Climbing specific maximum voluntary contraction (MVC) and time to task failure (TTF) of finger flexor muscles seem to be greater in male climbers than in male non-climbers (MacLeod et al. 2007; Quaine et al. 2003; Vigouroux and Quaine 2006; Ferguson and Brown 1997). The underlying physiological mechanisms may be enhanced vascular conductance and muscular re-oxygenation (MacLeod et al. 2007; Ferguson and Brown 1997), moreover, from a neurological point of view a slower decrease of the EMG power spectra (Vigouroux and Quaine 2006; Quaine et al. 2003) in male climbers compared to male non-climbers.

Gender differences in the performance of finger flexor muscles in climbers

Only one study compared female and male climbers (Mermier et al. 2000). They found that finger grip strength-to-weight ratio was significantly higher in male than female

climbers whereas finger pincer strength-to-weight ratio was not different. A continuous grip endurance test performed at 50% MVC revealed no difference in TTF between female and male climbers. In contrast, several authors studied differences in grip endurance between non-climbing women and men (Hunter et al. 2006, 2009; Thompson et al. 2007). MVC was higher in men than in women but TTF for a low-force continuous isometric fatiguing contraction was greater for women than for men whereas no difference in TTF was found for a high-force continuous isometric fatiguing contraction (Thompson et al. 2007; Hunter et al. 2006). Women had a significantly greater TTF than men for an intermittent isometric fatiguing contraction (Hunter et al. 2009). The physiological mechanisms behind those findings remain unclear as ultrasonography revealed no differences in blood flow between women and men (Thompson et al. 2007). Taken together, these observations indicate that a high-force fatiguing contraction provokes similar vascular occlusion in women and men whereas a low-force fatiguing contraction seems to be more favourable for women because of less vascular occlusion. Differences in skeletal muscle fatigue resistance between women and men may also result from different fibre type compositions (Wüst et al. 2008).

To the authors' knowledge, no study investigated specific finger flexor endurance in continuous and intermittent isometric tests and the related oxygenation of forearm muscles between female climbers and non-climbers.

We hypothesized that climbers have greater finger flexor MVC, a greater TTF and faster muscle re-oxygenation for an intermittent isometric endurance test than non-climbers and furthermore that female participants have lower finger flexor MVC but greater TTF for a continuous and an intermittent isometric contraction.

This study investigated the climbing specific finger flexor strength and endurance and related muscular oxygenation in female and male climbers and non-climbers.

Methods

Participants

Twelve climbers (6 males and 6 females) and 12 non-climbers (6 males and 6 females) volunteered to participate in the study. Characteristics of the study participants are shown in Table 1. The climbers were partly recruited from the Austrian National Climbing Team ($n = 6$) and partly from high-level student climbers ($n = 6$) of the Department of Sport Science of the University of Innsbruck. All non-climbers were sport students not involved in climbing. Climbing experience and climbing abilities have been recorded by interview. To analyse climbers' abilities

(Table 2) statistically, the French climbing grade scale was converted in a metric scale developed by Watts et al. (1993) (Table 3). Written consent was provided by every participant. The study was approved by the institutional review board.

Design

The test protocol was highly inspired by the work of MacLeod et al. (2007). The main difference was that every participant conducted the entire tests in one single test session in order to avoid day-to-day variability of strength and endurance patterns. Anthropometrical measurements were followed by a specific warm-up and familiarizing

with the testing device. After the maximal voluntary contraction (MVC) trials, the first endurance test was performed. This was followed by a recovery phase on a cycle ergometer and the second endurance test (Fig. 1).

Measurements

Anthropometric measurements and body composition

After clinical routine examinations, stature and body mass were assessed using standard techniques. A tetrapolar, multi-frequency bioelectrical impedance analysis (BIA 2000, Data Input GmbH, Germany) was used to determine percentage of body fat, lean body mass and body cell mass.

Table 1 Characteristics of study participants

	Climbers (mean $\pm$ SD)		Non-climbers (mean $\pm$ SD)		ANOVA results (<i>P</i> values)		
	Females (<i>n</i> = 6)	Males (<i>n</i> = 6)	Females (<i>n</i> = 6)	Males (<i>n</i> = 6)	Main effects		Interactions
					Climbers/non-climbers	Gender	Climbers/non-climbers $\times$ Gender
Age (years)	24.2 $\pm$ 2.14	26.7 $\pm$ 5.54	26.0 $\pm$ 1.27	26.8 $\pm$ 2.14	NS	NS	
Height (cm)	162.1 $\pm$ 2.60	179.2 $\pm$ 6.39	170.8 $\pm$ 7.03	178.2 $\pm$ 6.62	NS	0.001	NS
Body mass (kg)	54.1 $\pm$ 3.63	70.5 $\pm$ 9.79	63.5 $\pm$ 9.44	73.7 $\pm$ 6.39	NS	0.001	NS
Body mass index (kg/m ²)	20.6 $\pm$ 1.32	21.8 $\pm$ 1.58	21.7 $\pm$ 2.25	23.2 $\pm$ 0.99	NS	0.049	NS
Body fat mass (%)	20.3 $\pm$ 4.13	13.9 $\pm$ 2.61	25.9 $\pm$ 3.26	16.8 $\pm$ 2.20	0.003	0.001	NS
Lean body mass (%)	79.5 $\pm$ 3.95	86.1 $\pm$ 2.63	74.1 $\pm$ 3.27	83.2 $\pm$ 2.22	0.004	0.001	NS
Body cell mass (%)	46.2 $\pm$ 2.44	51.4 $\pm$ 1.43	41.7 $\pm$ 3.00	49.7 $\pm$ 1.84	0.003	0.001	NS

Table 2 Climbers' experience and abilities

	Female climbers (<i>n</i> = 6)			Male climbers (<i>n</i> = 6)			<i>t</i> test results (<i>P</i> values)
	Mean $\pm$ SD	Min.	Max.	Mean $\pm$ SD	Min.	Max.	
Experience (years)	14.0 $\pm$ 2.35	10	16	11.7 $\pm$ 4.08	7	17	0.289
Best on-sight (Watts scale)	3.71 $\pm$ 0.401	3	4	3.88 $\pm$ 0.647	3.25	4.75	0.603
Best lead (Watts scale)	4.42 $\pm$ 0.258	4	4.75	4.79 $\pm$ 0.749	4	5.75	0.289
Best bouldering (Watts scale)	3.66 $\pm$ 0.665	3	4.5	3.96 $\pm$ 0.621	3	4.75	0.336

Table 3 Climbing scale conversion chart by Watts et al. (1993)

French system	7b	7b+	7c	7c+	8a	8a+	8b	8b+	8c	8c+	9a	9a+
Numerical code	3.00	3.25	3.50	3.75	4.00	4.25	4.50	4.75	5.00	5.25	5.50	5.75

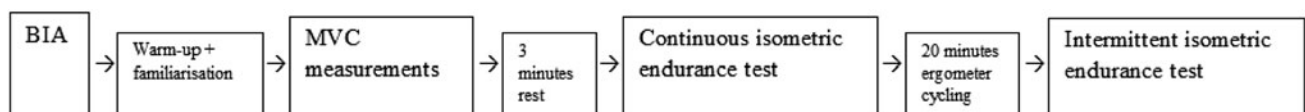


Fig. 1 Test protocol

Finger flexor strength and endurance

The testing apparatus to measure finger flexor MVC and finger flexor endurance was a modified version of an existing testing apparatus developed by Raschner et al. (2010). For comparability reasons to the study of MacLeod et al. (2007), the modifications consisted of extending the upper profiles of the testing apparatus and building a wooden ledge using the alloy profiles as rails (Fig. 2). A wire cable which had nearly no elasticity, fixed the ledge to a strain gauge (Hottinger Baldwin Messtechnik, Darmstadt, Germany). The ledge had a width of 22 mm and was covered by a grip tape. The wooden board, which formed the basis of the ledge, was fully adjustable in vertical direction but because it used the alloy profiles as vertical rails, no horizontal movement of the ledge was possible. Via a multi-channel analogue-to-digital converter (Spider8, Hottinger Baldwin Messtechnik, Darmstadt, Germany), data were collected, analysed and displayed in a Lab View program (National Instruments Corp., Austin, USA). The Lab View program was written for this study based on the software descriptions by MacLeod et al. (2007). Three test-modes were available: MVC, isometric continuous endurance and isometric intermittent endurance. During endurance testing, the participants got feedback of the applied force via a computer monitor by a “traffic lights” system. The green colour meant the applied force was correct, the blue light meant that the applied force was 5% or more over the target force and the red light meant that the applied force was 5% or more below the target force.

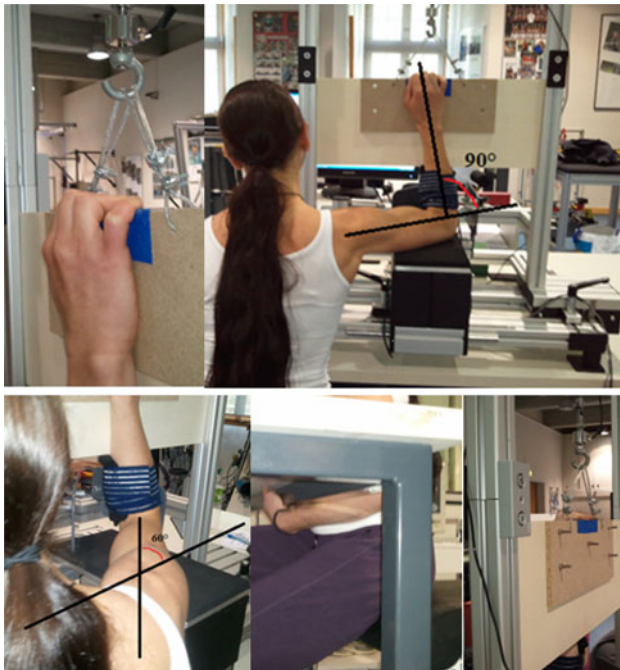


Fig. 2 Test apparatus with body and finger positioning

The test was automatically stopped if the applied force remained 5% below the target force for 2 s.

The participants were seated on a chair adjustable in height and as close as possible to the testing device (Fig. 2). The upper arm was in a horizontal position on the elbow rest and was adducted at 60° relative to the shoulder girdle axis. Upper arm and lower arm formed an angle of 90°. The participants were allowed to block the other arm under the table to ensure the correct position. The fingers of the dominant hand were placed in an open crimp position on the ledge. To ensure a climbing specific force transmission, only the distal phalanges of the index, middle, ring and little finger touched the ledge. The position could easily be controlled through the maintaining of the open crimp position. The thumb as well as the rest of the hand was not involved in any force application onto the ledge.

A standardized warm-up was performed to guarantee an optimal function of the finger flexor muscles. The first part of the warm-up consisted of several exercises, such as arm turning, gripping a 5 kg weight and circling the weight from one hand to the other around the body, pull-ups and arm hang. The second part of the warm-up was already on the finger flexor test device. The participants were properly seated, the grip kind and position was explained and the height of the ledge adjusted and noted. To get a specific warm-up and guarantee a familiarization to the ledge and the computer software, every participant performed 15 intermittent repetitions at 50% of the estimated MVC. This was followed by a simulation of the isometric continuous endurance test at 40% of the estimated MVC for 40 s. After that, isometric intermittent endurance test was simulated at the same intensity for five repetitions.

Each participant had four MVC trials separated by 30 s. Force had to be built up from zero to maximum force over 5 s. If the fourth trial was the best one, a fifth one was performed to be sure that best MVC was recorded. Participants were asked to contract as hard as possible on the ledge and were verbally encouraged.

The last MVC test and the continuous isometric finger flexor contraction were separated by a rest period of 3 min. This period was used to fix the near infrared spectroscopy (NIRS) probe to the underarm. The fixing spot was marked with a pen to assure that the probe was fixed at the same spot for the second endurance test. The target force was set at 40% of the best personal MVC. The clock started automatically once the target force was reached (green light). The participants were instructed to hold the target force as long as possible and were verbally encouraged.

Between the continuous endurance test and the intermittent endurance test, an active rest period of 20 min was set. Heyman et al. (2009) showed that cycling on an ergometer for 20 min at low intensity is a mean to preserve performance when repeating acute exhausting climbing

trials. Women were asked to cycle at 30 W, and men at 40 W (Ergo Bike 8008 TRS, Daum Electronic, Fürth, Germany). After the rest period, the ledge height was rearranged when necessary and the NIRS probe was placed on the same spot as in the first test. The target force for the intermittent isometric endurance test was exactly the same as in the first endurance test, but this time for every repetition the contraction phase lasted for 10 s followed by a 3 s rest. Participants were verbally encouraged to perform as many repetitions as possible.

Near infra-red spectroscopy (NIRS)

Changes in finger flexor muscle oxygenation were recorded using near infrared spectroscopy (Niro 200, Hamamatsu Photonics K.K., Hamamatsu City, Japan). The tissue oxygenation index ($\text{TOI} = \text{rSO}_2$) was chosen as master index for muscle oxygenation. Maximal de-saturation was calculated for the continuous isometric test. Re-oxygenation ($\Delta [\text{rSO}_2]$) was calculated for three time points of the intermittent isometric endurance test (TOI 1; TOI 2; TOI 3) and was expressed as mean $\Delta [\text{rSO}_2]$ of the first three 3-s rest phases, intermediate three 3-s rest phases and last three 3-s rest phases.

The emission and the detection probes were held by the original Hamamatsu probe holder, which allows a perfect positioning and a constant distance of 35 mm between the two probes. The pathlength (L) was calculated as follows: $L = \text{differential pathlength factor (DPF)} \times \text{distance between emitting probe and detecting probe in cm}$ (d) = $3.59 \times 3.5 = 12.57$ cm. The sampling rate was set at 6 Hz.

MacLeod et al. 2007 set the probes on the flexor digitorum superficialis. The difficult location of this muscle and the fact that the flexor digitorum profundus is more important in climbing because it bends the last joint of finger 2–5, made us chose a different location for the probes. To get the best results, we placed the probes by drawing an imagined straight line from the carpus to the medial epicondyle of the humerus. Each participant did one contraction on the ledge of the testing device to locate the area with the greatest contraction on the imagined line where the probe was to be fixed.

Statistical analysis

Data are presented as mean $\pm$ SD. Differences in group means regarding climbing abilities were assessed using two-sampled t test. Differences in the group means regarding anthropometric, strength, endurance and NIRS data were assessed using two-way ANOVA (main effects: climbers/non-climbers, gender; interaction: climbers/non-climbers $\times$ gender). Correlation analyses were applied to

examine a possible relation between two variables. Statistical significance level was set at $P < 0.05$. The force-time integral for the endurance tests were calculated to get an expression for “climbing specific” endurance because this integral takes into count the different target forces (MacLeod et al. 2007). Data analyses were conducted with the SPSS statistical software package (Version 17.0).

Results

Characteristics of study participants are shown in Table 1. There were no differences between groups with regard to age. Climbers had a significantly higher percentage of lean body mass ($P = 0.004$) and body cell mass ($P = 0.003$) and a significantly lower percentage of body fat mass ($P = 0.003$) than non-climbers. Female and male participants differed significantly with regard to height ($P < 0.001$), body mass, body mass index ($P = 0.049$), percentage of body fat mass ($P < 0.001$), percentage of lean body mass ($P < 0.001$), and percentage of body cell mass ($P < 0.001$).

In women, on-sight climbing ability ranged from 7b to 8a and best lead climbing performance ranged from 8a to 8b+. In men, on-sight climbing ability ranged from 7b+ to 8b+ and best lead climbing performance ranged from 8a to 9a+. There were no significant differences between female and male climbers' group means regarding experience, the on-sight climbing performance, the best lead climbing performance and the best bouldering performance.

Force and endurance characteristics are presented in Table 4. MVC and strength-to-weight ratio (MVC/body mass) were significantly greater for climbers than for non-climbers ($P = 0.003$; $P < 0.001$) and significantly greater for men than for women ($P < 0.001$; $P = 0.002$). TTF for the intermittent test and the force-time integrals for the continuous and the intermittent test were significantly greater for climbers than for non-climbers ($P = 0.030$; $P = 0.027$; $P = 0.005$). Men showed a significantly greater force-time integral for the continuous test than women ($P = 0.017$). No interactions between climbers/non-climbers and gender could be found for all force and endurance parameters. Close correlations were demonstrated between the best on-sight climbing performance and strength-to-weight ratio ($r^2 = 0.946$, $P < 0.001$) and between the best lead climbing performance and strength-to-weight ratio ($r^2 = 0.839$, $P = 0.010$) in female climbers.

A typical TOI curve of the continuous endurance test is presented in Fig. 3. During the intermittent test, the TOI curve had a specific pattern that was distinguished by a fall in O_2 saturation during the 10 s of contraction and a rise in O_2 saturation during the three seconds of rest. De-saturation neither differed between climbers and non-climbers nor between women and men for the continuous isometric

Table 4 Gender-specific strength and endurance characteristics of finger flexor muscles in climbers and non-climbers

	Climbers (mean $\pm$ SD)		Non-climbers (mean $\pm$ SD)		ANOVA results (<i>P</i> values)		
	Females (<i>n</i> = 6)	Males (<i>n</i> = 6)	Females (<i>n</i> = 6)	Males (<i>n</i> = 6)	Main effects		Interactions
					Climbers/ non-climbers	Gender	Climbers/ non-climbers $\times$ gender
MVC (N)	311.6 $\pm$ 56.89	491.0 $\pm$ 76.82	227.6 $\pm$ 31.75	402.2 $\pm$ 74.15	0.003	0.001	NS
MVC/body mass (N/kg)	5.8 $\pm$ 1.27	7.1 $\pm$ 1.31	3.6 $\pm$ 0.61	5.4 $\pm$ 0.64	0.001	0.002	NS
Continuous TTF (s)	132.0 $\pm$ 53.98	92.6 $\pm$ 31.16	117.2 $\pm$ 16.39	100.9 $\pm$ 25.47	NS	NS	
Intermittent TTF (s)	452.5 $\pm$ 240.96	295.2 $\pm$ 243.46	204.5 $\pm$ 97.94	194.3 $\pm$ 83.77	0.030	NS	NS
Force–continuous test time integral (N $\times$ s)	15,609 $\pm$ 4,819.4	17,433 $\pm$ 2,878.6	10,523 $\pm$ 7,09.0	15,929 $\pm$ 3,737.2	0.027	0.017	NS
Force–intermittent test time integral (N $\times$ s)	53,023 $\pm$ 25,048.2	52,760 $\pm$ 33,731.3	18,042 $\pm$ 7,459.1	30,688 $\pm$ 13,628.0	0.005	NS	NS

MVC maximum voluntary contraction, TTF time to task failure

Table 5 Tissue oxygenation index (TOI) during the continuous and intermittent finger flexor endurance test. TOI de-sat. continuous test means: maximum de-saturation during the continuous isometric endurance test

	Climbers (mean $\pm$ SD)		Non-climbers (mean $\pm$ SD)		ANOVA results (<i>P</i> values)		
	Females (<i>n</i> = 6)	Males (<i>n</i> = 6)	Females (<i>n</i> = 6)	Males (<i>n</i> = 6)	Main effect		Interactions
					Climbers/non-climbers	Gender	Climbers/non-climbers $\times$ gender
TOI de-sat. continuous test (%)	35.5 $\pm$ 14.93	40.2 $\pm$ 10.25	40.5 $\pm$ 15.71	34.2 $\pm$ 14.12	NS	NS	
TOI 1 int. test (%)	11.9 $\pm$ 4.90	18.8 $\pm$ 4.62	14.42 $\pm$ 5.48	11.1 $\pm$ 2.94	NS	NS	
TOI 2 int. test (%)	21.6 $\pm$ 5.67	24.3 $\pm$ 5.93	16.9 $\pm$ 8.02	14.3 $\pm$ 6.31	0.012	NS	NS
TOI 3 int. test (%)	24.8 $\pm$ 10.43	26.4 $\pm$ 6.34	18.1 $\pm$ 9.00	16.0 $\pm$ 4.46	0.015	NS	NS
TOI mean int. test (%)	19.5 $\pm$ 5.75	23.2 $\pm$ 4.93	16.5 $\pm$ 7.41	13.8 $\pm$ 3.77	0.014	NS	NS

TOI 1 int. test rest phase re-oxygenation of the first three 3-s rest phases

TOI 2 int. test rest phase re-oxygenation of the intermediate three 3-s rest phases

TOI 3 int. test rest phase re-oxygenation of the last three 3-s rest phases

endurance test. The intermittent isometric endurance test revealed a significantly faster re-oxygenation for TOI 2 ($P = 0.012$), TOI 3 ($P = 0.015$) and TOI mean ($P = 0.014$) for climbers compared to non-climbers. No differences in re-oxygenation for the intermittent isometric endurance test were found between women and men. No interactions were observed between climbers/non-climbers and gender with regard to de-saturation and re-oxygenation in the continuous and intermittent isometric endurance tests (Table 5).

Discussion

The main findings of this study are first that climbers, regardless of gender, had significantly greater endurance

performance of finger flexor muscles associated with a longer TTF for an intermittent isometric endurance test compared to non-climbers. Second, in elite female climbers, the best on-sight and lead climbing performance can be explained by the strength-to-weight ratio.

Using a similar test device and a nearly identical test protocol as MacLeod et al. (2007), we found not only a significantly greater finger flexor force-time integral but also a significantly greater finger flexor TTF for the intermittent isometric endurance test for climbers than for non-climbers. This may be explained by the extraordinary high climbing abilities of the participants in the present study. Other studies investigating elite climbers found similar results using different testing devices and test protocols (Vigouroux and Quaine 2006; Quaine et al. 2003; Ferguson and Brown 1997). However, none of these studies included

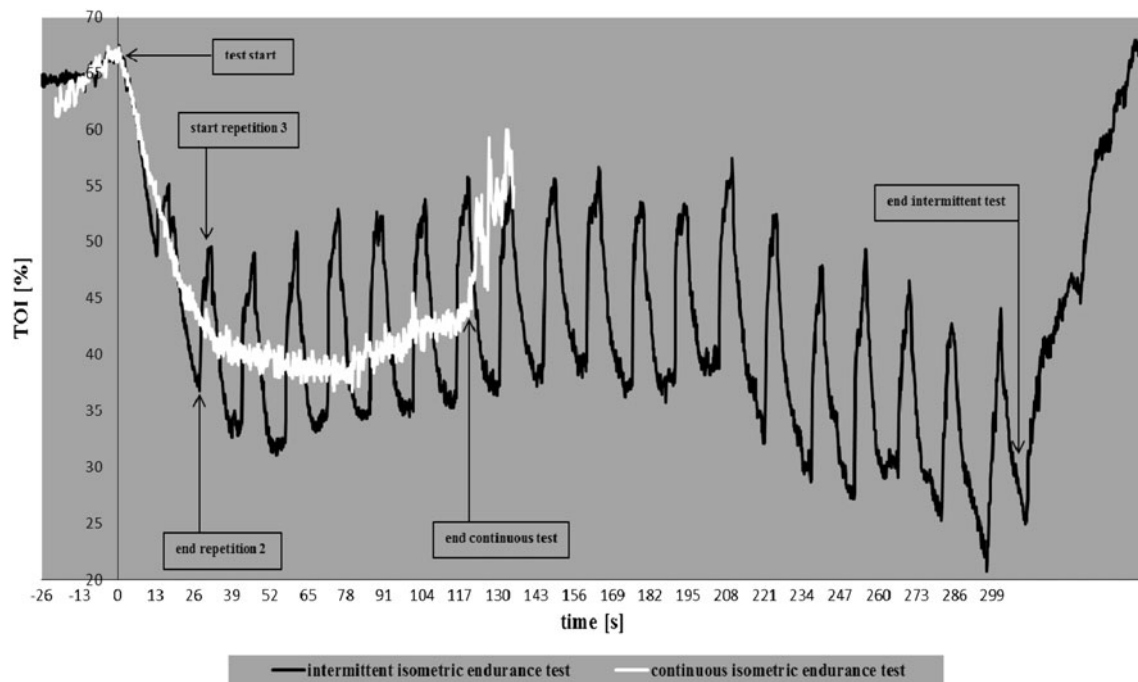


Fig. 3 Example of a near infrared spectroscopy trace (NIRS) trace from the continuous and the intermittent isometric endurance test. Both traces are from the same participant. The *x* axis represents time

in seconds, 0 marks the test beginning. The *y* axis represents the tissue oxygenation index (TOI)

female climbers. We could show that female and male climbers did not differ with regard to intermittent finger flexor endurance when compared to female and male non-climbers. We thus conclude that male as well as female elite climbers with long term climbing experience may benefit from the same training effects regarding climbing specific finger flexor endurance.

A positive relationship between the best on-sight and lead climbing performance and finger flexor strength-to-weight ratio was found for female climbers. This is a common finding and has also been reported by other researchers (Wall et al. 2004; Mermier et al. 2000). However, the close relationships between the best on-sight climbing performance and strength-to-weight ratio, and the best lead climbing performance and strength-to-weight ratio in this study are surprising. This may well be due to the homogenous elite group of female climbers. All female climbers were active world-cup climbers or former successful competitive climbers. It should be pointed out that the strength-to-weight ratio was influenced only to a small extent by body mass and BMI (range ± 6.7 and 6.4%) but to a large extent by strength (range $\pm 18.3\%$). The superior intermittent finger flexor endurance of climbers over non-climbers may be explained by the faster re-oxygenation of the finger flexor muscles observed in climbers compared to non-climbers during short rest phases. As demonstrated by MacLeod et al. (2007), rest phase re-oxygenation for the time points 1 and 2 was significantly faster in climbers than non-climbers. Furthermore,

mean re-oxygenation of the 3 time points was significantly faster in climbers than non-climbers. The time course of re-oxygenation differed neither between female and male climbers nor between female and male non-climbers. This is an important observation because this is the first study including female top climbers. Forearm muscle re-oxygenation seems to be a highly trainable factor. Specific adaptations with regard to muscle perfusion may play a role (Ferguson and Brown 1997). Alomari et al. (2010) showed that 4 weeks of rhythmic handgrip training lead to a significant elevation of forearm blood flow. Muscle perfusion partly depends on capillary-to-fibre ratio and capillary density. Resistance exercise, intense intermittent endurance training and moderate continuous endurance training have been shown to improve capillary-to-fibre ratio and capillary density (Robbins et al. 2009; Jensen et al. 2004; Green et al. 1999; McCall et al. 1996; Henriksson 1992; Klausen et al. 1981). Regarding forearm muscles, climbing is a mixture of resistance exercise and intense intermittent endurance training. Adaptations in capillary-to-fibre ratio and capillary density may therefore be induced by climbing. The major benefit of a faster re-oxygenation during short rest phases is the faster re-synthesis of phosphocreatine (McMahon and Jenkins 2002; Tomlin and Wenger 2001).

The intermittent isometric endurance test revealed no significant differences in finger flexor TTF between women and men. This result differs from the results of Hunter et al. (2009) who found that women with lower finger flexor

MVC had a significantly greater TTF than men. These differences might be due to the use of different muscle groups, different testing devices and different test protocols compared to the present study.

In the present study, climbing specific finger flexor MVC and strength-to-weight ratio was significantly greater for climbers than non-climbers. These findings are in line with the findings of several other studies showing that climbers have greater climbing specific finger flexor MVC and strength-to-weight ratio than non-climbers (MacLeod et al. 2007; Vigouroux and Quaine 2006; Quaine et al. 2003; Grant et al. 2001) but in contradiction to the findings of Ferguson and Brown (1997). Findings regarding MVC and strength-to-weight ratio always depend on climbers' level as well as anthropometric, physical activity and strength characteristics of the control group. If not for MVC, a climbing specific test setup should at least show advantages for climbers regarding strength-to-weight ratio. We assume that the test setup used in the present study may be considered as climbing specific. In this study, the continuous isometric endurance test revealed no significant differences in TTF between women and men. Thompson et al. (2007) highlighted the importance of the percentage of MVC at which a continuous isometric test is performed by showing that TTF for a continuous isometric test always depends on differences in MVC and the chosen target force. A low-force fatiguing contraction may provide physiological benefits for those with low MVC in terms of muscle perfusion and oxygenation. In our test situation, the target force of 40% of MVC seems to be too high for women to benefit from their lower MVC.

The continuous test revealed no significant difference in muscle de-saturation between climbers and non-climbers probably indicating that during continuous isometric contraction, where energy is largely produced anaerobically, limits of de-saturation do not differ between trained and less trained muscles. Other studies showed that trained forearm muscles may benefit from a better oxygenation than untrained forearm muscles when performing continuous isometric fatiguing contractions (Usaj et al. 2007; Usaj 2001, 2002). Because target force in these studies was equal for every participant (150 N) the results are not comparable to the results of the current study where target force was chosen relative to MVC (40% MVC).

The main limitations may arise from the test setup and test design applied in this study. Despite the fact that this test setup has been described to be climbing-specific (MacLeod et al. 2007), it has to be considered that gripping feels different in real climbing. Furthermore, despite the use of an active rest between the first and the second test as proposed by Heyman et al. (2009), we cannot entirely exclude that the first test affected the second one. Nevertheless, pre-tests with a small sample size indicated that the

chosen recovery method was likely to guaranty that the first test did not affect the second test in a negative way.

Conclusion

The major finding of this study is that intermittent isometric endurance of the finger flexor muscles is significantly greater in climbers than non-climbers without differences between sexes. This advantage is associated with faster re-oxygenation during short rest phases between the isometric contraction phases and may at least partly result from specific long-term training. In addition, strength-to-weight ratio of the finger flexor muscles was found to be an excellent predictor for best on-sight and lead climbing performance in world-class female climbers.

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